



ROLL NO

PUNJAB PUBLIC SERVICE COMMISSION
COMBINED COMPETITIVE EXAMINATION
FOR RECRUITMENT TO THE POSTS OF
PROVINCIAL MANAGEMENT SERVICE, ETC -2023
CASE NO. 1C2024

SUBJECT: PHYSICS (PAPER-I)

TIME ALLOWED: THREE HOURS

MAXIMUM MARKS: 100

NOTE:

- i. All the parts (if any) of each Question must be attempted at one place instead of at different places.
- ii. Write Q. No. in the Answer Book in accordance with Q. No. in the Q. Paper.
- iii. No Page/Space be left blank between the answers. All the blank pages of Answer Book must be crossed.
- iv. Extra attempt of any question or any part of the question will not be considered.

NOTE: Attempt any FIVE Questions in All. Attempt in Urdu or English.

- Q. No.1** **a)** What is conical pendulum? Give its two uses. Derive a relation for its period of revolution.
b) Define and explain angular momentum in detail. Derive a correlation between angular momentum and torque.
(10+10=20 Marks)
- Q. No.2** **(a)** State and prove Kepler's law of periods for planetary motion.
(b) Define and explain moment of inertia. Calculate the rotational inertia of a solid sphere about its diameter.
(8+12=20 Marks)
- Q. No.3** **a)** How do fluid properties like viscosity, surface tension, and elasticity affect fluid flow and fluid motion?
b) Water is flowing through a pipe with a diameter of 10 cm. The pipe narrows down to a diameter of 5 cm at a constriction. If the pressure in the wide section is 100 kPa and the velocity of the water is 2 m/s, what is the pressure in the narrow section of the pipe if the water is incompressible and there is no energy loss due to friction?
(12+8=20 Marks)
- Q. No.4** **a)** An upward thrust is used to lift the rocket. Explain this phenomenon analytically and mathematically. Derive a relationship to find this upward thrust using Bernoulli's equation.
b) What is Poiseuille's law? Derive mathematical formulation of this law.
(8+12=20 Marks)
- Q. No.5** **(a)** What is an interferometer, and how does it work in the context of observing Newton's rings?
(b) How can the phase difference between the interfering waves be measured using Newton's rings, and what is its significance?
(10+10=20 Marks)
- Q. No.6** **(a)** What are Lissajous figures and how are they formed?
(b) Explain what is meant by free oscillation with two degrees of freedom. Give an example and discuss the factors that affect the natural frequencies of the system.
(10+10=20 Marks)
- Q. No.7** **(a)** Explain distinctive features of Maxwellian distribution of molecular velocities.
(b) Differentiate between Bose-Einstein and Fermi-Dirac statistics.
(10+10=20 Marks)
- Q. No.8** **a)** Differentiate between a heat engine and refrigerator on the basis of their construction, working principle and efficiencies.
b) What do you understand by low temperature physics? Describe different mechanism of production of low temperatures.
(10+10=20 Marks)



ROLL NO

PUNJAB PUBLIC SERVICE COMMISSION
COMBINED COMPETITIVE EXAMINATION
FOR RECRUITMENT TO THE POSTS OF
PROVINCIAL MANAGEMENT SERVICE, ETC -2023
CASE NO. 1C2024

SUBJECT: PHYSICS (PAPER-II)

TIME ALLOWED: THREE HOURS**MAXIMUM MARKS: 100****NOTE:**

- i. All the parts (if any) of each Question must be attempted at one place instead of at different places.
- ii. Write Q. No. in the Answer Book in accordance with Q. No. in the Q. Paper.
- iii. No Page/Space be left blank between the answers. All the blank pages of Answer Book must be crossed.
- iv. Extra attempt of any question or any part of the question will not be considered.

NOTE: **Attempt Five Questions in All. Calculator is allowed (Non-Programmable). Attempt in Urdu or English.**

- Q. No. 1:** a) State and Prove Poynting Theorem.
b) Write down FOUR Maxell's equations.
c) What is Poynting Vector. Write down its Unit. **(10+4+6=20 Marks)**
- Q. No. 2:** a) Explain the formation of Hysteresis Loop in a Ferromagnetic Material. What is Coercivity and Retentivity.
b) An electron is circulating in a circular orbit of radius 5.1 Pm . If a magnetic field of induction of $B = 2.00 \text{ webers/m}^2$ acts at right angle to the plane of the orbit, Calculate the change in its magnetic moment. **(10+10=20 Marks)**
- Q. No. 3:** a) Define Fermi level and sketch energy band diagram for a p-n junction diode.
b) Discuss the characteristics and I-V curve for a diode in forward and reverse biased conditions.
c) Explain the construction and working of a light emitting diode (LED). **(4+10+6=20 Marks)**
- Q. No. 4:** a) What are the advantages of digital electronics over analogue? Describe major limitations to the use of digital techniques.
b) What are the advantages of octal over hexadecimal number system? Why is hexadecimal number system used instead of the octal number system when working with 8- and 16-bit digital computers?
c) How error correction is done using parity codes? **(8+8+4=20 Marks)**
- Q. No. 5:** a) What is Pauli Exclusion Principle. How it is helpful for the distribution of electrons in different orbits of an atom.
b) What are Quantum Numbers. Explain their importance. **(12+8=20 Marks)**
- Q. No. 6:** a) What is Zeeman effect. Explain Anomalous Zeeman effect.
b) Calculate the energy of d-orbital of the 3rd orbit of H-atom under the influence of weak external magnetic field. **(12+8=20 Marks)**
- Q. No. 7:** a) What is a mass spectrograph? Discuss its construction, working principle, operation and applications.
b) What do you mean by nuclear spin? How nuclear magneton is different from Bohr magneton?
c) Differentiate between natural and artificial sources of radioactivity. **(8+8+4=20 Marks)**
- Q. No. 8:** a) Explain the phenomena of nuclear fusion in detail. Elaborate the nuclear fusion reactions taking place on sun.
b) Write a brief note on the construction of a TOKAMAK. **(15+5=20 Marks)**